# A Posteriori Error Estimation

Residuals, local indicators, and reliability

# From a rate to an error estimate

Chapter 7 ended with an a priori statement such as

$$\|u - u_h\|_a \leq C_I \sqrt{k_1}\, h^{s-1}\, |u|_{H^s(\Omega)}\,.$$

It predicts a rate, but it does not answer two questions after a computation:

1. Is the error on this mesh small enough?
2. Which cells should be refined?

Chapter 8 replaces the unknown derivative of $u$ by defects computed from $u_h$:

$$\boxed{\text{solve} \longrightarrow \text{estimate} \longrightarrow \text{mark} \longrightarrow \text{refine.}}$$

Today: derive the residual, localize it to cells and facets, and build the estimator. Marking and refinement follow in the next class.

# The residual is what the discrete solution leaves unsatisfied

For the diffusion problem, let

$$a(w, v) = \int_\Omega k \nabla w \cdot \nabla v \, \mathrm{d}x, \qquad \ell(v) = \int_\Omega f v \, \mathrm{d}x - \int_{\Gamma_N} q v \, \mathrm{d}s.$$

The exact and discrete problems are

$$a(u, v) = \ell(v) \quad \forall v \in \mathcal{V}_0, \qquad a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in \mathcal{V}_{h,0}.$$

Define the residual functional

$$\boxed{R(u_h; v) = \ell(v) - a(u_h, v), \qquad v \in \mathcal{V}_0.}$$

With $e = u - u_h$,

$$\boxed{R(u_h; v) = a(e, v).}$$

Therefore $R(u_h; v_h) = 0$ for every $v_h \in \mathcal{V}_{h,0}$: the residual vanishes on the test space used to compute $u_h$, but generally not on all of $\mathcal{V}_0$.

# In the energy norm, residual size equals error size

Because $a(\cdot,\cdot)$ is an inner product,

$$|R(u_h; v)| = |a(e, v)| \leq \|e\|_a \, \|v\|_a .$$

Taking the supremum gives one inequality. Taking $v = e$ gives equality:

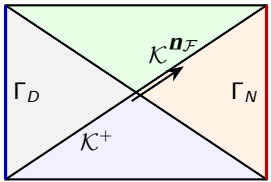
$$\boxed{\|e\|_a = \sup_{0 \neq v \in \mathcal{V}_0} \frac{|R(u_h; v)|}{\|v\|_a}.}$$

This identity is exact, but it is not yet computable:

- the maximizing direction is $v = e$, which is unknown;
- the supremum ranges over the infinite-dimensional space $\mathcal{V}_0$;
- the single number does not yet say where the error is located.

The next step rewrites $R(u_h; v)$ as cell and facet contributions.

# A conforming mesh has three facet classes




- $\mathcal{F}_h^{\mathrm{int}}$: shared by two cells $\mathcal{K}^+$ and $\mathcal{K}^-$;
- $\mathcal{F}_h^D$: boundary facets on $\Gamma_D$;
- $\mathcal{F}_h^N$: boundary facets on $\Gamma_N$.

Choose $\boldsymbol{n}_\mathcal{F}$ from $\mathcal{K}^+$ to $\mathcal{K}^-$. For a piecewise quantity $w$,

$$[\![w]\!]_\mathcal{F} = w^+ - w^-.$$

Reversing the cell labels reverses both the jump and the normal, so the normal-flux jump used below is unchanged.

## Cellwise integration by parts exposes the strong defect

Start from

$$R(u_h; v) = \int_\Omega fv \, \mathrm{d}x - \int_{\Gamma_N} qv \, \mathrm{d}s - \int_\Omega k\nabla u_h \cdot \nabla v \, \mathrm{d}x.$$

On one cell $\mathcal{K}_e$,

$$\int_{\mathcal{K}_e} k\nabla u_h \cdot \nabla v \, \mathrm{d}x = -\int_{\mathcal{K}_e} \nabla\cdot(k\nabla u_h)v \, \mathrm{d}x + \int_{\partial\mathcal{K}_e} (k\nabla u_h \cdot \boldsymbol{n}_e)v \, \mathrm{d}s.$$

After summing over cells,

$$\begin{aligned} R(u_h; v) = &\sum_e \int_{\mathcal{K}_e} \big(f + \nabla\cdot(k\nabla u_h)\big)v \, \mathrm{d}x \\ &- \sum_e \int_{\partial\mathcal{K}_e} (k\nabla u_h \cdot \boldsymbol{n}_e)v \, \mathrm{d}s - \int_{\Gamma_N} qv \, \mathrm{d}s. \end{aligned}$$

The first line is the elementwise balance defect. The second line must be regrouped facet by facet.

# Facet regrouping gives two flux defects

Define the computed flux $\boldsymbol{q}_h = -k\nabla u_h$. On an interior facet, the two cell-boundary terms combine into

$$j_{\mathcal{F}} = -[\![k\nabla u_h \cdot \boldsymbol{n}_{\mathcal{F}}]\!]_{\mathcal{F}} = [\![\boldsymbol{q}_h \cdot \boldsymbol{n}_{\mathcal{F}}]\!]_{\mathcal{F}}.$$

On a Neumann facet,

$$j_{\mathcal{F}} = -\big(q + k\nabla u_h \cdot \boldsymbol{n}\big) = \boldsymbol{q}_h \cdot \boldsymbol{n} - q.$$

On $\Gamma_D$, $v = 0$, so there is no residual term. With

$$r_e = f + \nabla\cdot\left(k\nabla u_h\right) = f - \nabla\cdot \boldsymbol{q}_h,$$

the exact representation is

$$\boxed{R(u_h; v) = \sum_e \int_{\mathcal{K}_e} r_e v \,\mathrm{d}x + \sum_{\mathcal{F}\in\mathcal{F}_h^{\mathrm{int}}\cup\mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} v \,\mathrm{d}s.}$$

# What the residuals measure

**Cell residual**
$r_e = f - \nabla \cdot \boldsymbol{q}_h$
balance defect inside a cell

**Interior facet**
$j_{\mathcal{F}} = [\![\boldsymbol{q}_h \cdot \boldsymbol{n}_{\mathcal{F}}]\!]$
flux leaving one cell does not enter the next

**Neumann facet**
$j_{\mathcal{F}} = \boldsymbol{q}_h \cdot \boldsymbol{n} - q$
computed flux misses prescribed flux

For $p = 1$ and cellwise-constant $k$, $\nabla u_h$ is constant on each cell, so $\nabla \cdot (k \nabla u_h) = 0$ and $r_e = f$. The computed gradient information then appears in the facet jumps.

Galerkin orthogonality balances the two kinds of defects on every free basis function $\phi_i$:

$$\sum_e \int_{\mathcal{K}_e} r_e \phi_i \, \mathrm{d}x + \sum_{\mathcal{F}} \int_{\mathcal{F}} j_{\mathcal{F}} \phi_i \, \mathrm{d}s = 0.$$

Neither residual must vanish separately.

# Activity: residuals for the uniformly loaded bar

Let $L = AE = f = 1$, $\overline{P} = 0$, and use a uniform mesh of size $h$ with linear elements. The exact solution is

$$u(x) = x - \frac{x^2}{2},$$

and Chapter 7 showed $u_h = \mathcal{I}_h u$.

Work with a neighbor. For element $\mathcal{K}_e = (x_e, x_{e+1})$:

1. Find $u_h'|_{\mathcal{K}_e}$ using the chord slope of $u$.
2. Compute

$$r_e = f + (AEu_h')', \quad j_i = (AEu_h')(x_i^+) - (AEu_h')(x_i^-), \quad j_{N+1} = \overline{P} - (AEu_h')(L^-).$$

3. For an interior hat function $\phi_i$, verify that its element and facet residual contributions cancel.

The nodal values are exact. Predict: do all residuals vanish?

# Activity debrief: exact nodes do not imply zero residual

The chord slope equals the exact derivative at the midpoint $\bar{x}_e$:

$$u_h'|_{\mathcal{K}_e} = u'(\bar{x}_e) = 1 - \bar{x}_e.$$

Therefore

$$\boxed{r_e = 1, \qquad j_i = -h, \qquad j_{N+1} = -\frac{h}{2}.}$$

Every residual is nonzero, although $u_h(x_i) = u(x_i)$ at every node.

For an interior hat function,

$$\int_0^1 r\phi_i \, \mathrm{d}x = h, \qquad j_i\phi_i(x_i) = -h.$$

At the last node, the two terms are $h/2$ and $-h/2$. This is residual orthogonality in local form: the discrete equations balance cell and facet defects, but the defects still detect the error between nodes.

# Galerkin orthogonality supplies the powers of $h$

The residual representation alone contains $\|e\|_{L^2(\mathcal{K}_e)}$, which gives no local mesh-size weight. For any $v_h \in \mathcal{V}_{h,0}$,

$$R(u_h; v) = R(u_h; v - v_h).$$

Choose a local approximation $v_h = \mathcal{Q}_h v$. Hypothesis (Q) requires

$$\|v - \mathcal{Q}_h v\|_{L^2(\mathcal{K}_e)} \le C_Q \, h_e \, |v|_{H^1(\omega_e)} ,$$
$$\|v - \mathcal{Q}_h v\|_{L^2(\mathcal{F})} \le C_Q \, h_{\mathcal{F}}^{1/2} \, |v|_{H^1(\omega_{\mathcal{F}})} .$$

Here $\omega_e$ and $\omega_{\mathcal{F}}$ are fixed-size patches. The operator maps $\mathcal{V}_0$ into $\mathcal{V}_{h,0}$ and uses local averages, so it is defined for $H^1$ functions even when nodal interpolation is not.

The factors $h_e$ and $h_{\mathcal{F}}^{1/2}$ become the estimator weights.

# Local indicators assign a computable number to each cell

For each cell $\mathcal{K}_e$, define

$$\eta_e^2 = h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} + \frac{1}{2} \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{\mathrm{int}}}} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} + \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{N}}} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} \, .$$

The global estimator is

$$\eta = \left( \sum_e \eta_e^2 \right)^{1/2} .$$

The factor $1/2$ assigns half of an interior facet to each adjacent cell. After summing over cells, every facet is counted exactly once. A Neumann facet belongs to only one cell, so it receives no factor $1/2$.

# In one dimension, reliability has an explicit constant

For the bar with variable $AE(x) \geq (AE)_0 > 0$, nodal interpolation is defined for every $H^1$ error. The nodal terms vanish for $w = e - \mathcal{I}_h e$, leaving only element residuals. The interpolation bound and Cauchy–Schwarz give

$$\boxed{\|u - u_h\|_a \leq \frac{1}{\sqrt{(AE)_0}} \left( \sum_e h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} \right)^{1/2}.}$$

For the activity, $(AE)_0 = 1$, $r_e = 1$, $N = 1/h$, and

$$\eta_{1\mathrm{D}}^2 = Nh^3 = h^2, \qquad \eta_{1\mathrm{D}} = h.$$

Chapter 7 gave $\|e\|_a = h/\sqrt{12}$, so

$$I_{\mathrm{eff}} = \frac{\eta_{1\mathrm{D}}}{\|e\|_a} = \sqrt{12} \approx 3.46.$$

The estimator decreases at the correct rate but overestimates by a fixed factor on this mesh sequence.

# Multidimensional reliability: statement and proof route

Assume Hypothesis (Q), shape regularity, and $k \geq k_0 > 0$. Then

$$\boxed{\|u - u_h\|_a \leq C_{\mathrm{rel}}\eta, \qquad C_{\mathrm{rel}} = C_Q\sqrt{\frac{N_{\mathrm{ov}}}{k_0}}.}$$

The proof uses four steps:

1. $\|e\|_a^2 = R(u_h; e) = R(u_h; e - \mathcal{Q}_h e)$;
2. insert the cell-and-facet residual representation;
3. apply Hypothesis (Q), then Cauchy–Schwarz to the combined sums;
4. use finite patch overlap and $|e|_{H^1(\Omega)} \leq \|e\|_a / \sqrt{k_0}$.

The constants are independent of $h$, but $C_Q$ and $N_{\mathrm{ov}}$ are not normally computed. Thus $\eta$ controls the error up to a fixed factor; its displayed value is not automatically a certified numerical upper bound.

# Definition, reliability, efficiency, and effectivity differ

| Term | Claim |
|---|---|
| Estimator | $\eta$ and $\eta_e$ are computable definitions; a definition alone does not compare them with the error. |
| Reliability | $\|e\|_a \leq C_{\mathrm{rel}}\eta$. A small estimator implies a small error. |
| Local efficiency | $\eta_e \leq C_{\mathrm{eff}}(\|e\|_{a,\omega_e} + \mathrm{osc}_e)$. A large indicator should correspond to error on a nearby patch, up to data oscillation. |
| Effectivity | $I_{\mathrm{eff}} = \eta / \|e\|_a$, evaluated when the exact error is known, measures the numerical scale of the estimator. |

Today we proved the one-dimensional reliability result and stated the multidimensional result under Hypothesis (Q). Efficiency is a desired property and a definition here; no efficiency theorem is being claimed for this candidate estimator without a matching source and precise oscillation definition.

## What to retain and what comes next

- The residual is the error measured through the bilinear form: $R(u_h; v) = a(u - u_h, v)$.
- Cellwise integration by parts separates element balance defects from interior and Neumann flux defects.
- Galerkin orthogonality allows subtraction of a local approximation; this produces the powers of the local mesh size.
- $\eta_e$ localizes the estimator and $\eta$ combines the cells. Reliability means the estimator controls the energy error up to a mesh-independent constant.

**Exit question.** For linear elements with constant $k$ and $f = 0$ on every cell, can the estimator still be nonzero? Identify the term that remains.

**Next class.** Complete the multidimensional proof, compare marking strategies, and connect the indicators to conforming local refinement.